LAG-AWARE ADVISORY DIGITAL TWIN FOR DIAGNOSING BLAST-FURNACE PRODUCTION SHORTFALLS

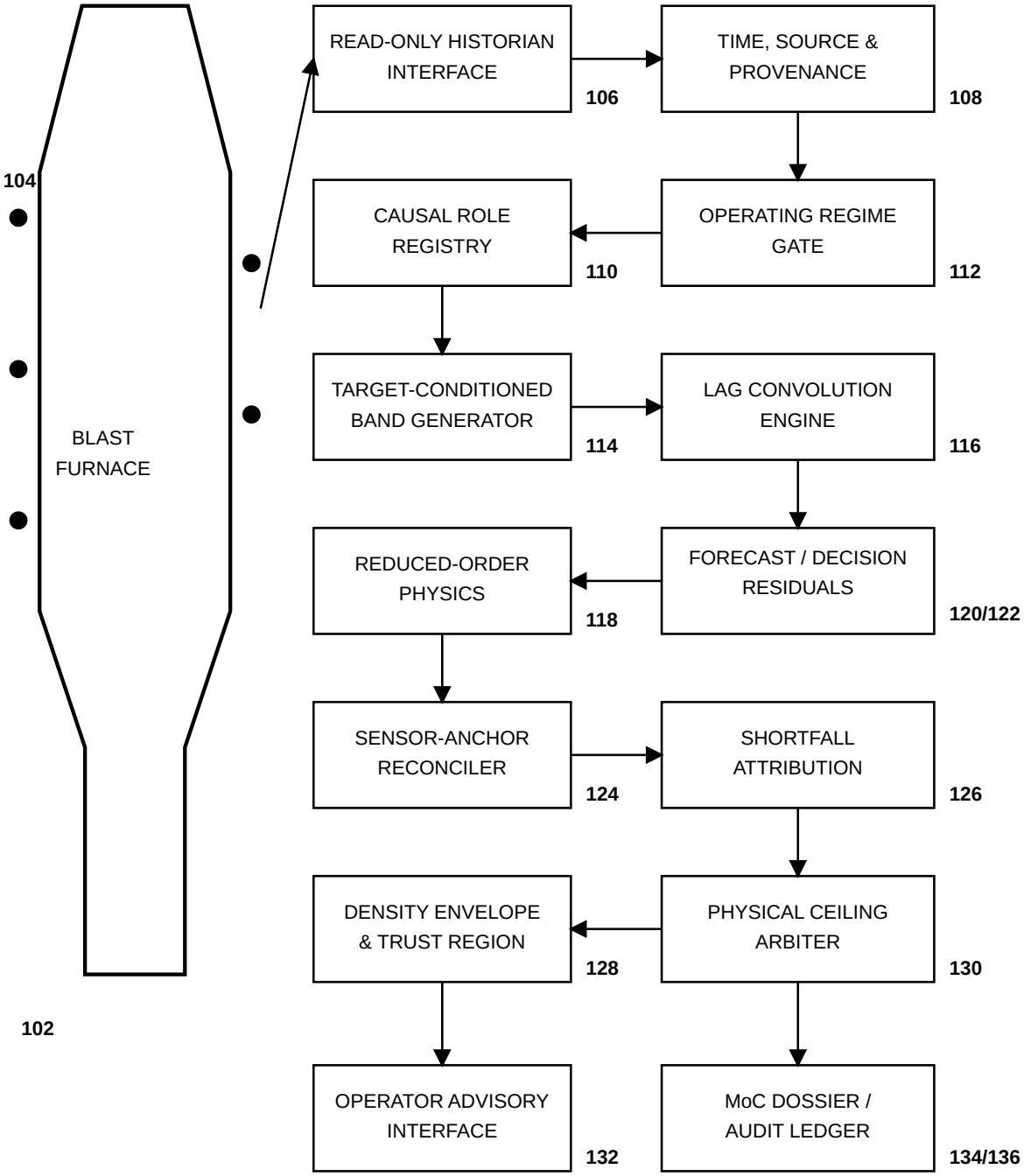


FIG. 1

SHORTFALL-DIAGNOSIS METHOD

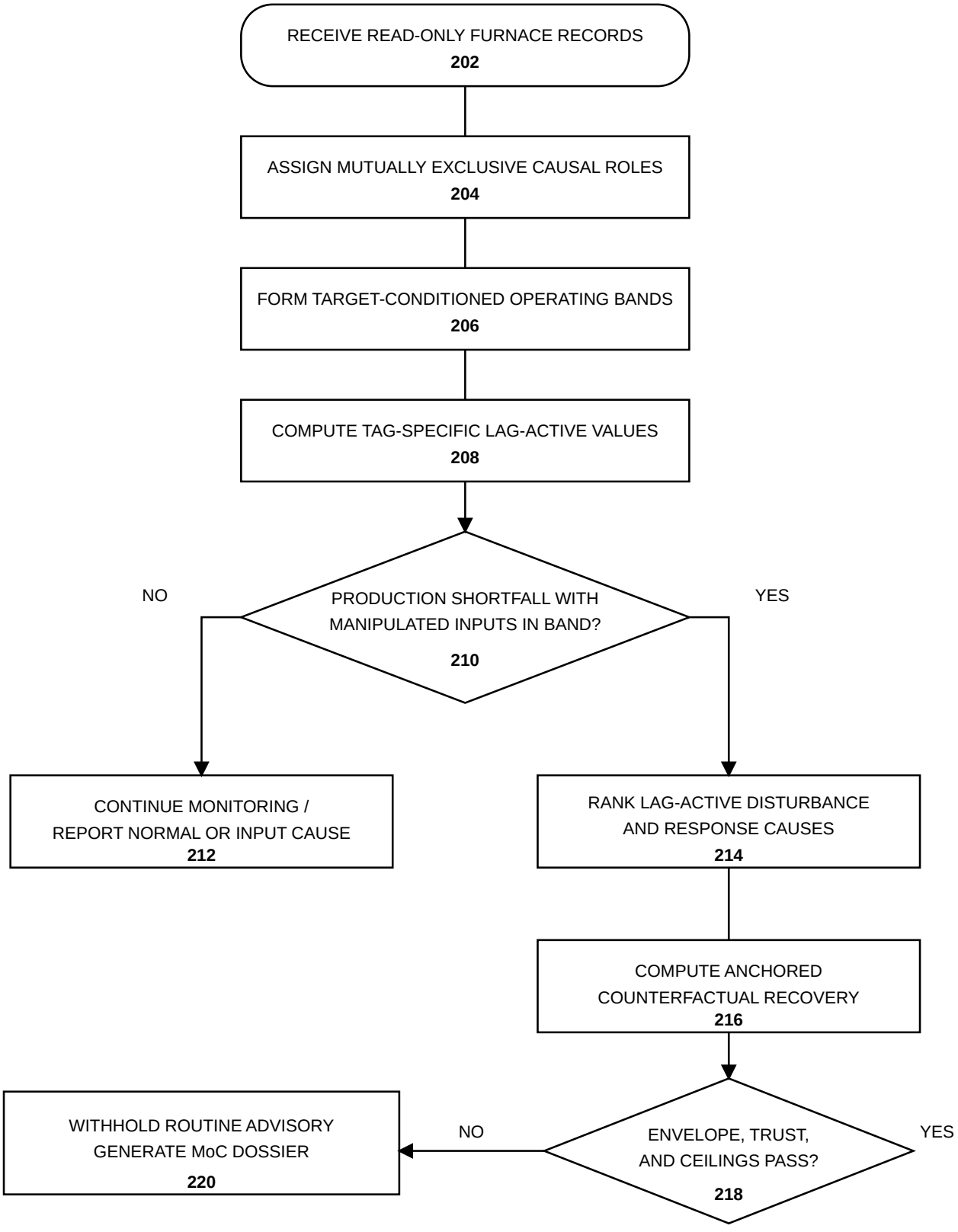
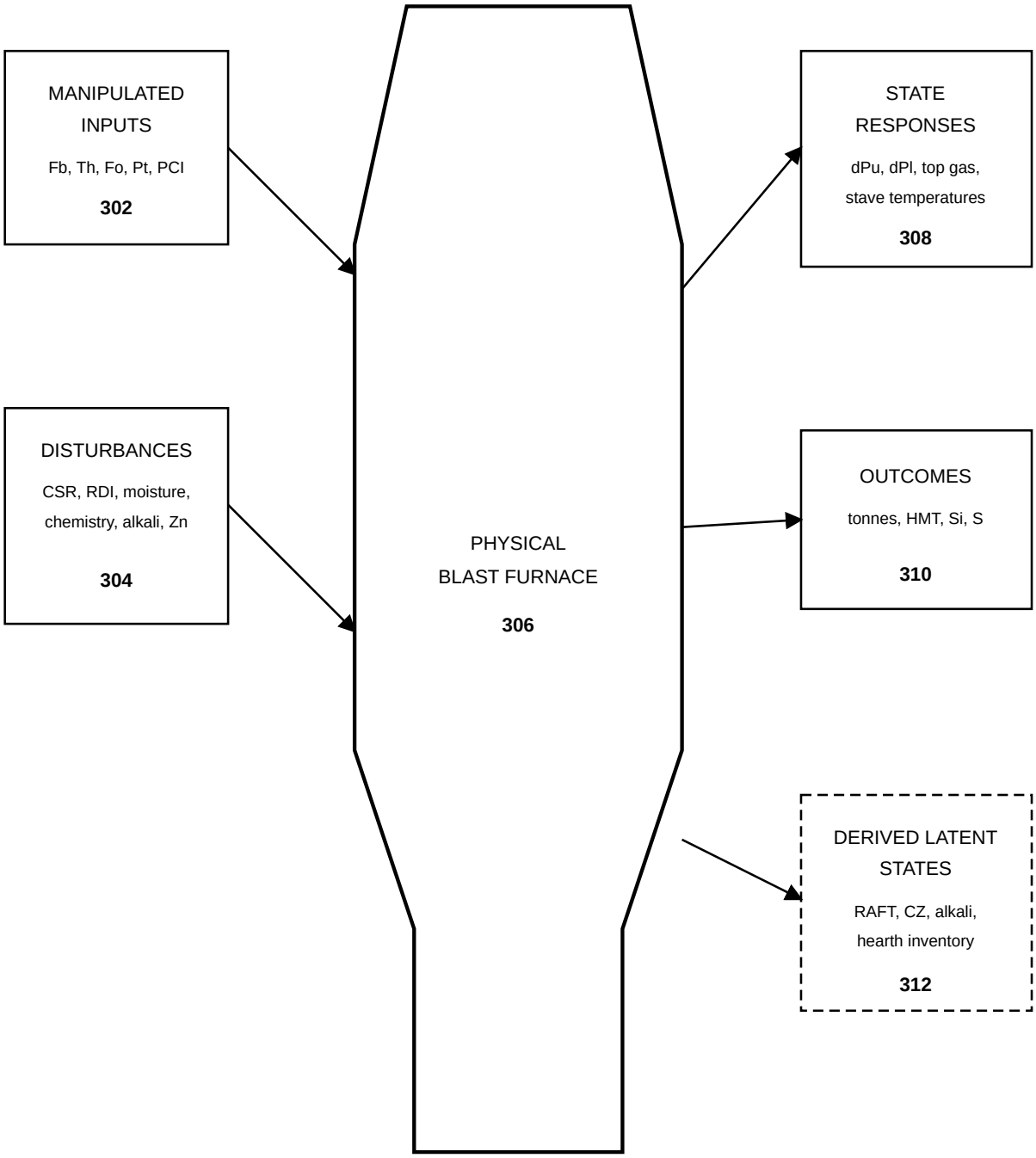


FIG. 2

CAUSAL-ROLE PARTITION AND FURNACE FLOW



EACH TAG HAS EXACTLY ONE ROLE

FIG. 3

TARGET-PRODUCTION-CONDITIONED OPERATING BANDS

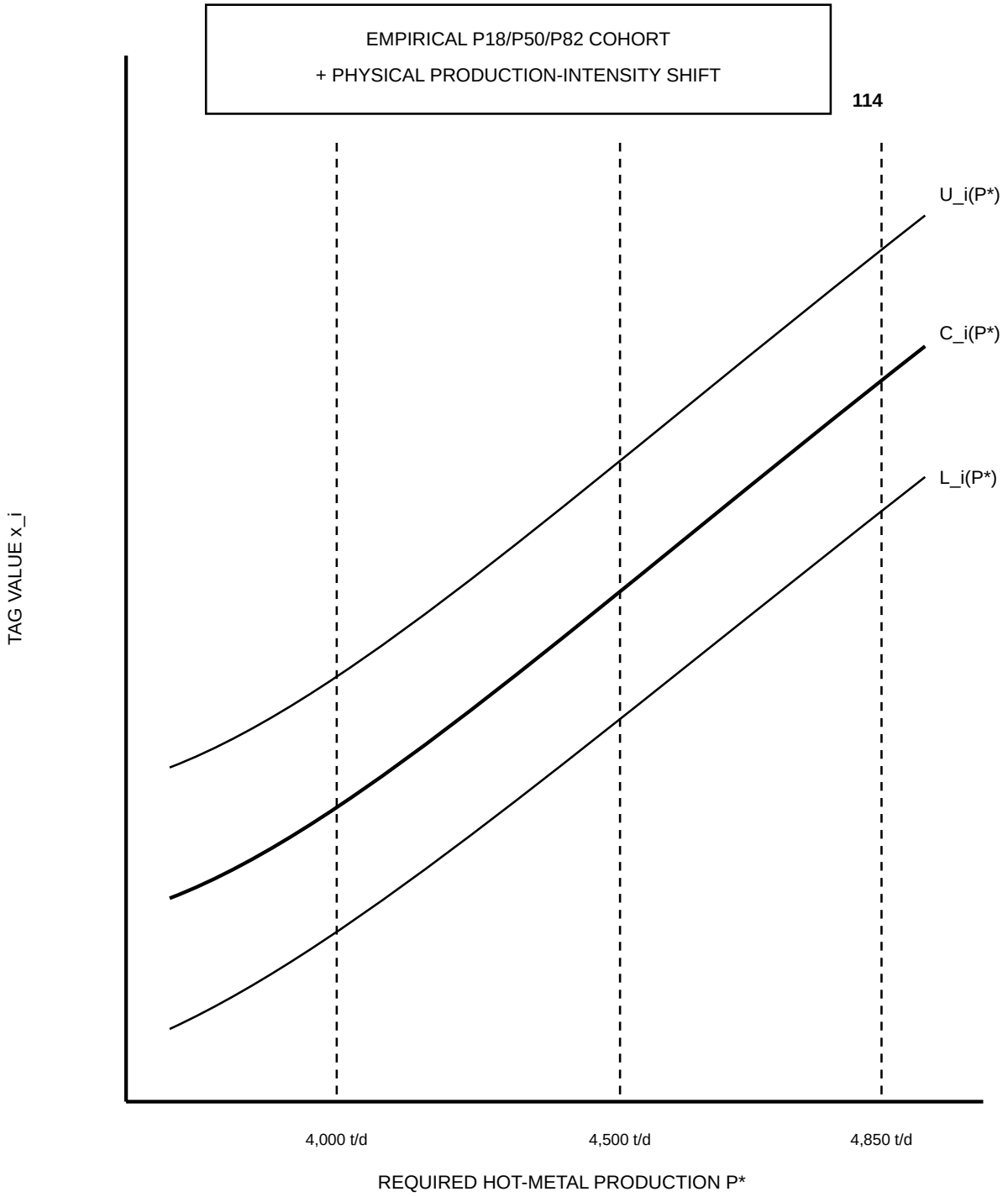
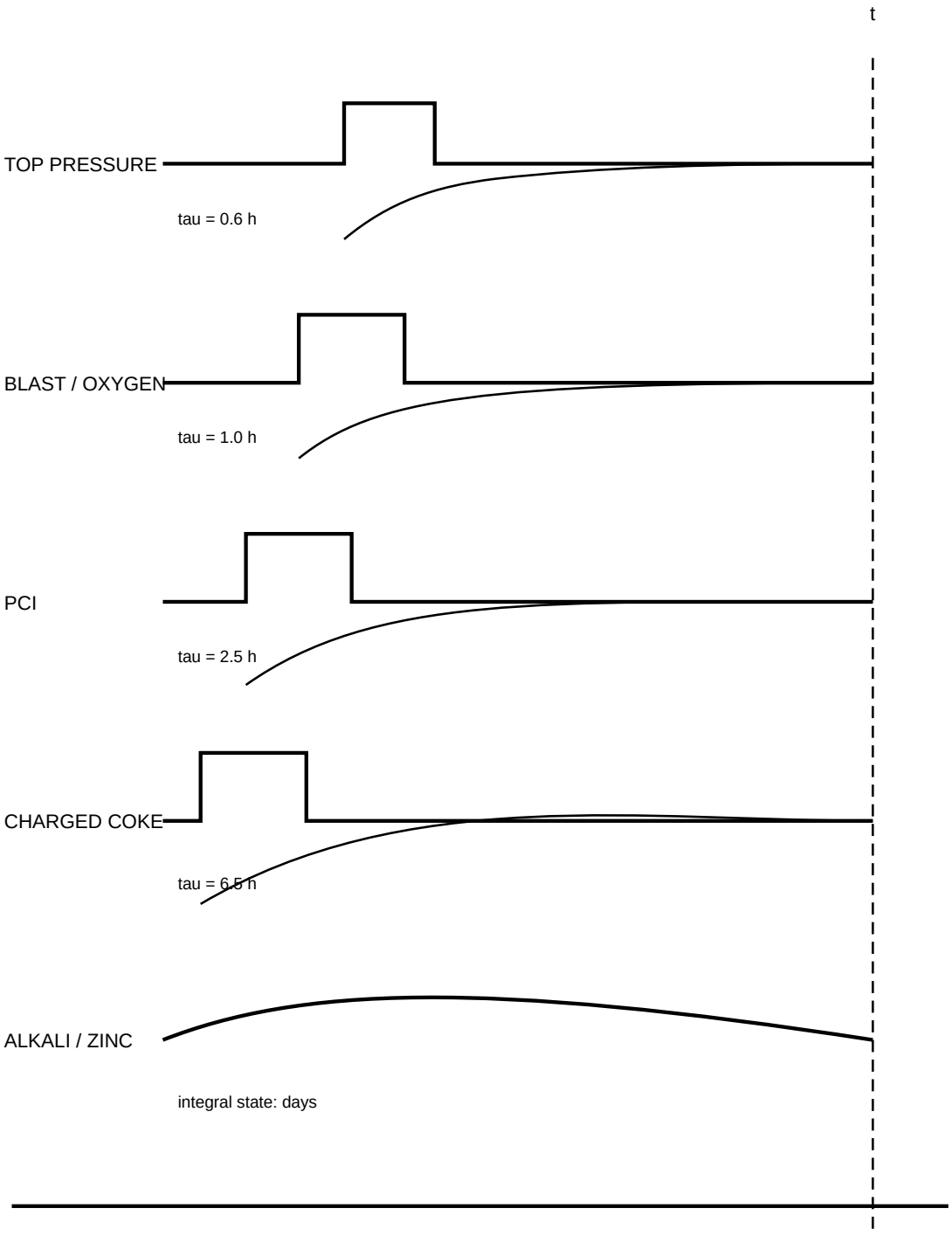


FIG. 4

TAG-SPECIFIC PHYSICAL LAG ALIGNMENT



THICK: INPUT HISTORY THIN: LAG-ACTIVE VALUE x_tilde_i(t)

FIG. 5

SENSOR-ANCHORED HYBRID TWIN AND ZERO-CHANGE IDENTITY

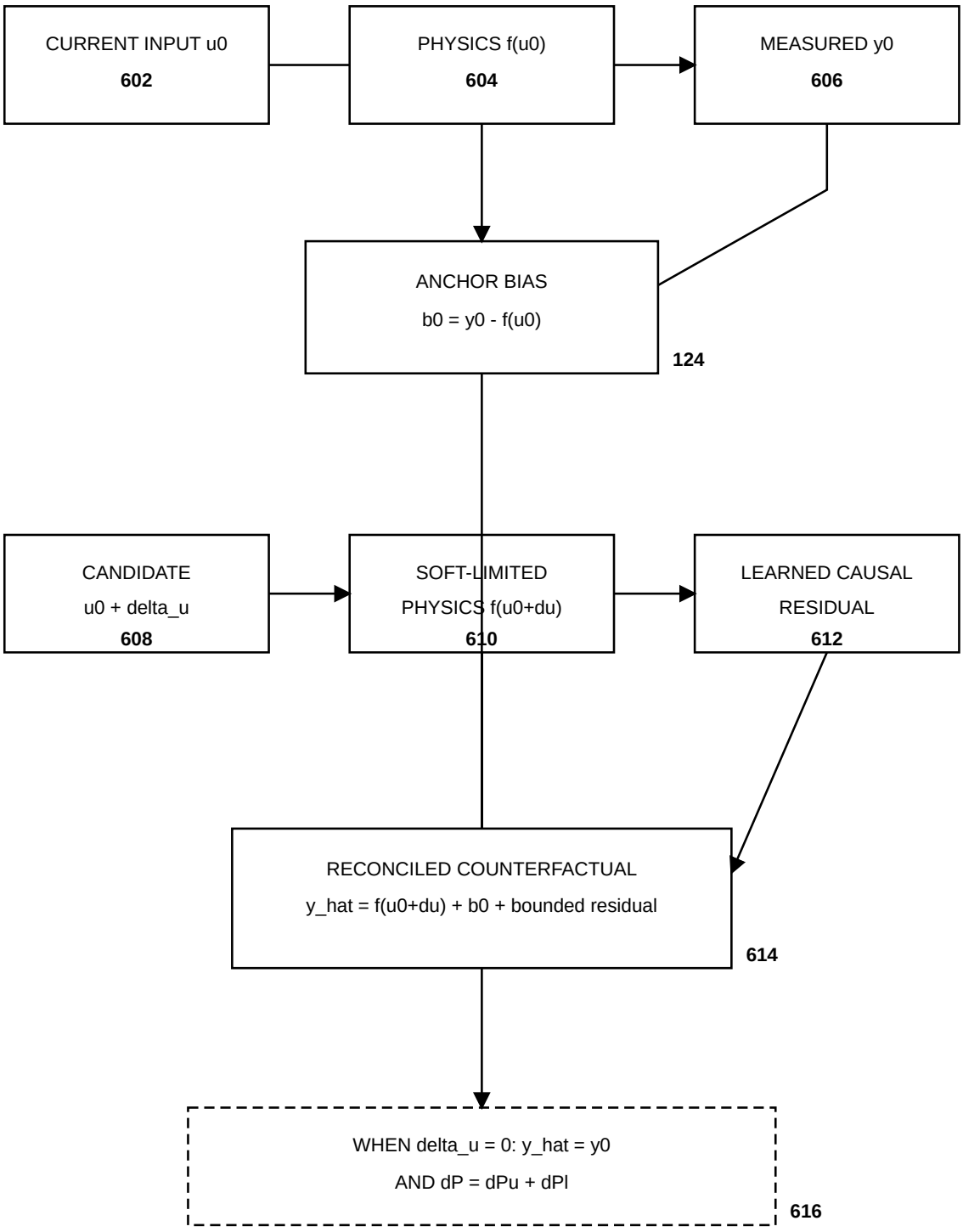
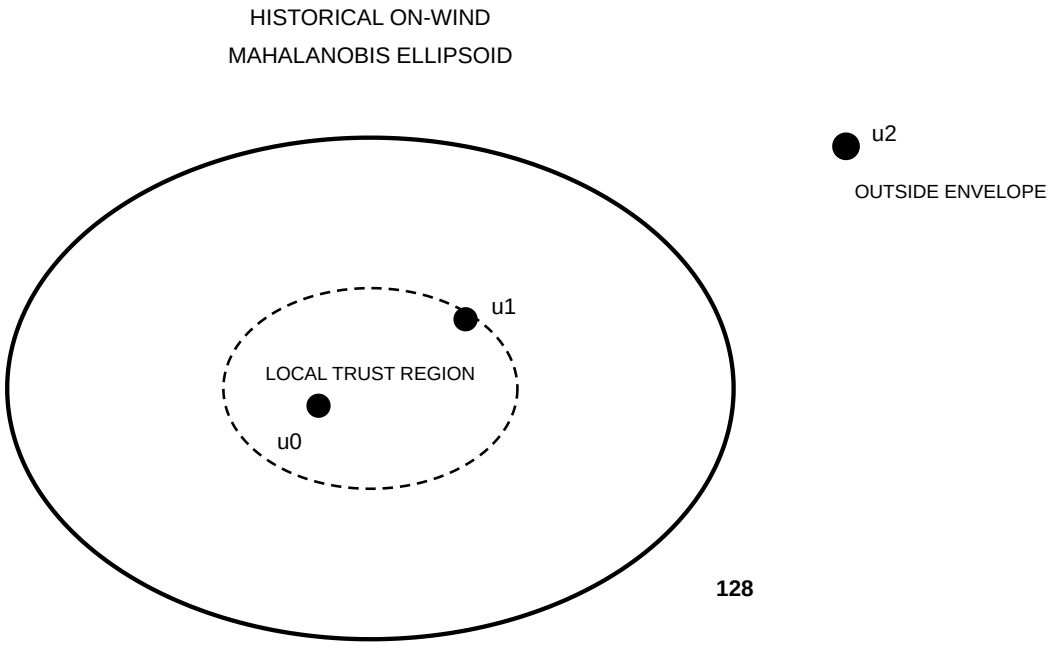


FIG. 6

JOINT-DENSITY ENVELOPE, TRUST REGION, AND CEILING ARBITER



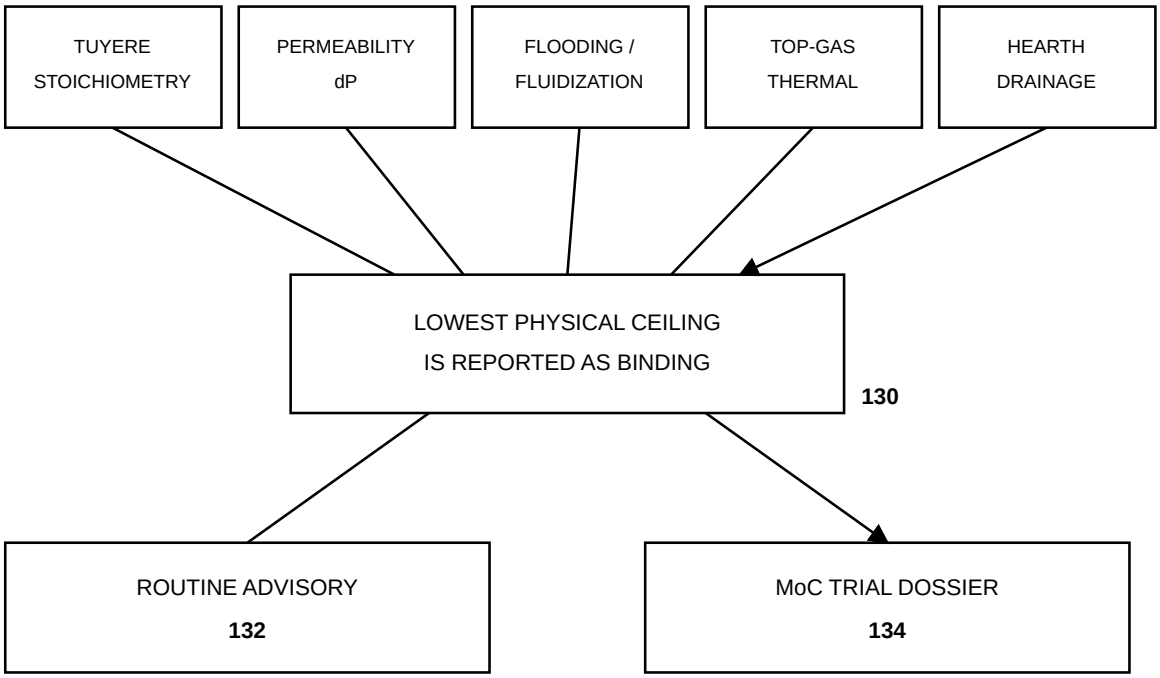


FIG. 7

TAMPER-EVIDENT ADVISORY AND SEPARATE EXECUTION CONFIRMATION

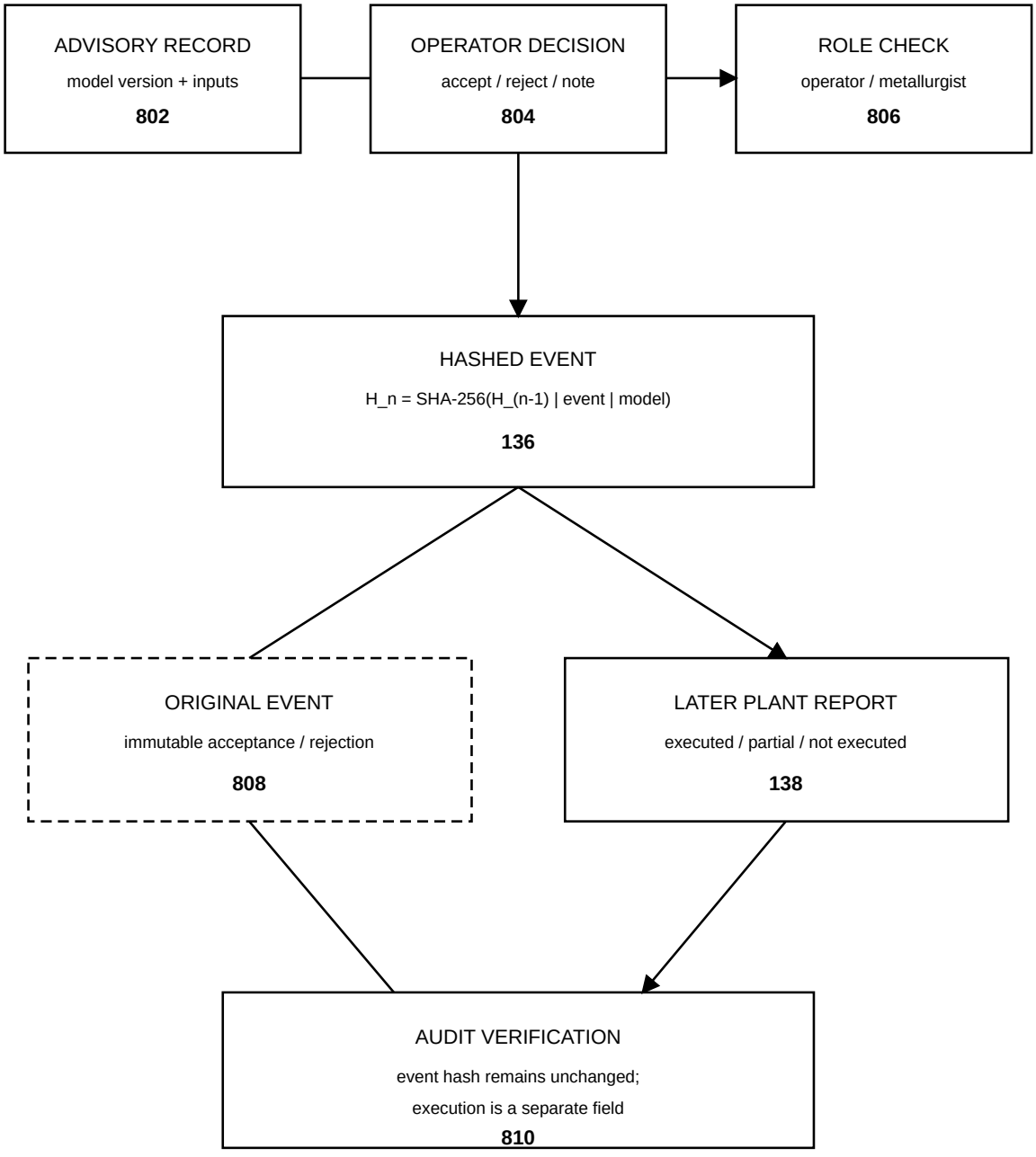


FIG. 8